

Nutrient Profiles of Vegetarian and Non Vegetarian Dietary Patterns



Background Differences in nutrient profiles between vegetarian and non vegetarian dietary patterns reflect nutritional differences that may contribute to the development of disease. Objective To compare nutrient intakes between dietary patterns characterized by consumption or exclusion of meat and dairy products. Design Cross-sectional study of 71751 subjects (mean age 59 years) from the Adventist-Health-Study-2. Data was collected between 2002 and 2007. Participants completed a 204-item validated semi-quantitative food frequency questionnaire. Dietary patterns compared were: non vegetarian, semi vegetarian, pesco vegetarian, lacto-ovo vegetarian and strict vegetarian. ANCOVA was used to analyze differences in nutrient intakes by dietary patterns and were adjusted for age, and sex and race. BMI and other relevant demographic data were reported and compared by dietary pattern using chi-square tests and ANOVA. Results Many nutrient intakes varied significantly between dietary patterns. Non vegetarians had the lowest intakes of plant proteins, fiber, β -Carotene, and Mg than those following vegetarian dietary patterns and the highest intakes of saturated, trans, arachidonic, and docosahexaenoic fatty acids. The lower tails of some nutrient distributions in strict vegetarians suggested inadequate intakes by a portion of the subjects. Energy intake was similar among dietary patterns at close to 2000 kcal/d with the exception of semi vegetarians that had an intake of 1713 kcal/d. Mean BMI was highest in non-vegetarians (mean; standard deviation [SD]) (28.7; [6.4]) and lowest in strict vegetarians (24.0; [4.8]). Conclusions Nutrient profiles varied markedly between dietary patterns that were defined by meat and dairy intakes. These differences can be of interest in the etiology of obesity and chronic diseases.